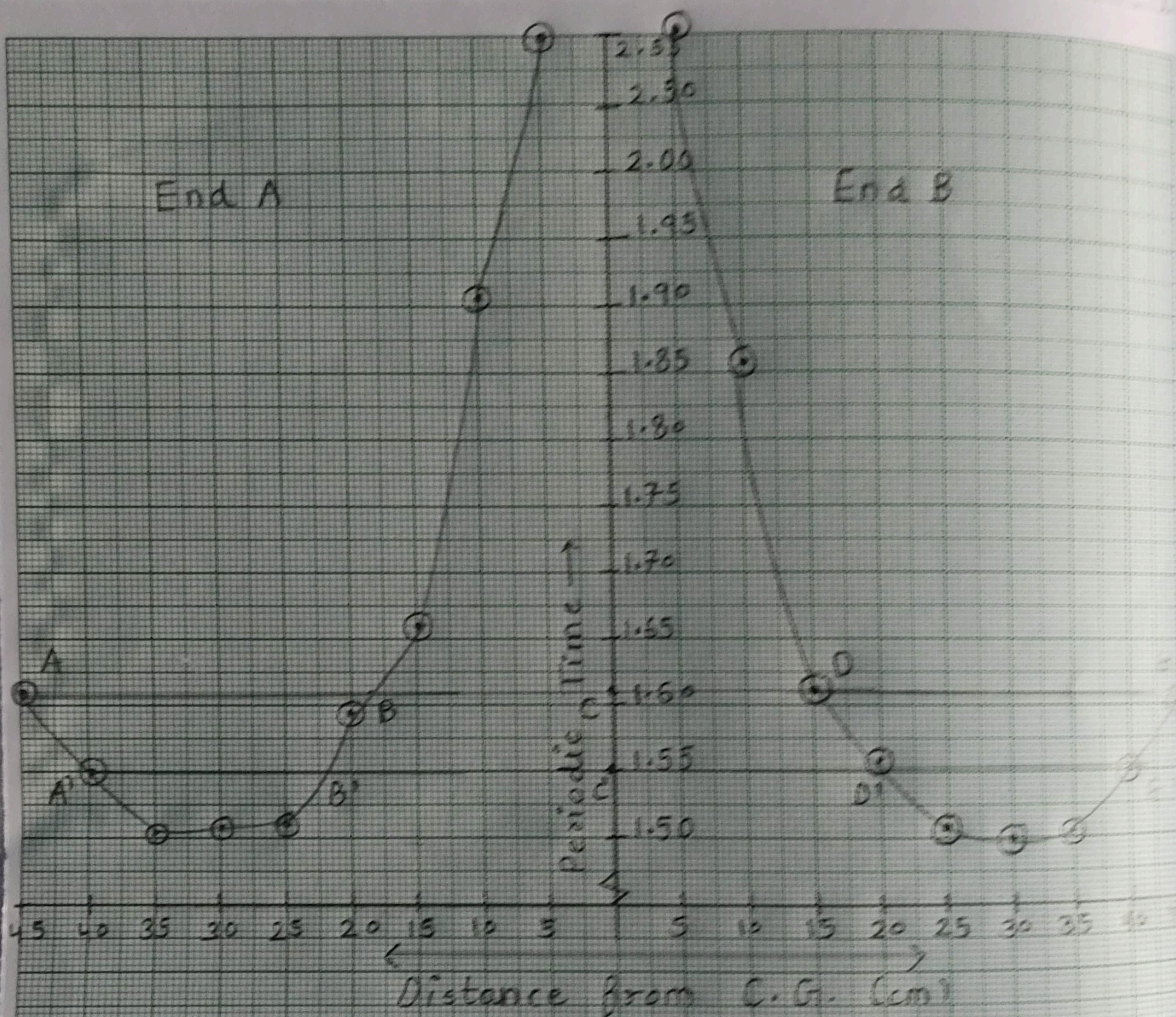




- * Aim : (1) To determine the value of acceleration due to gravity ' g ' by a bar pendulum.
(2) To find the radius of gyration of the pendulum.

* Apparatus : Bar pendulum, spirit level (for leveling the brackets), telescope, stop watch, meter-rod, a rigid support fixed on the wall.

* Theory : The bar is allowed to oscillate about a horizontal knife edge passing through each of holes in turn, beginning with that nearest to one end and the time period is noted in each case. If a graph is plotted between the distance of the knife edges from the centre of gravity and the corresponding time period T , it has the shape shown in graph.

If a horizontal line $ABCDE$ is drawn, it cuts the graph in points A, B, O, E about which the time period is the same.

For A as the centre of suspension, D is the centre of oscillation and for E as the centre of suspension, B is the centre of oscillation.

The distance AD and BE thus give the length of the equivalent simple pendulum L . If the corresponding time period is T , then

Observations:

Least count of stop watch = 0.01 s.

Least count of meter scale = 0.1 cm.

S. No.	Distance of the hole from G (cm)	On Side A		On Side B	
		Time for 20 vibs (s)	Time Period	Time for 20 vibs (s)	Time Period
1.	45	32.45	1.622	32.10	1.605
2.	40	31.00	1.55	30.88	1.544
3.	35	30.10	1.505	30.05	1.502
4.	30	30.16	1.508	30.00	1.50
5.	25	30.35	1.517	30.26	1.513
6.	20	31.80	1.59	31.28	1.564
7.	15	33.41	1.670	32.31	1.615
8.	10	38.19	1.909	37.06	1.853
9.	5	52.44	2.622	51.06	2.553

Calculations:

(i) From the line ABCDE

$$l_1 = \frac{AC + CE}{2}$$

2

$$l_2 = \frac{BC + CD}{2}$$

2

Equivalent length

$$l = l_1 + l_2 = 62.5 \text{ cm}$$

$$T = 1.62 \text{ sec}$$

$$g = \frac{4\pi^2 l}{T^2} = 941.75 \text{ cm s}^{-2}$$

$$K_1 = \sqrt{l_1 l_2} = 117.39 \text{ cm}$$

$$T = 2\pi \sqrt{\frac{L}{g}}$$

Or

$$g = \frac{4\pi^2 L}{T^2}$$

The radius of gyration : $K = \sqrt{l_1 l_2}$

here $l_1 = \frac{AC+CE}{2}$ and $l_2 = \frac{BC+CD}{2}$

- * Procedure :
- (1) Fix the knife edges in the outermost holes on either end of the bar parallel to each other with their sharp edges pointing towards G.
 - (2) Check the levelling of the bracket with a spirit level.
 - (3) Suspend the pendulum by placing the knife edge on the side A perpendicular to the slot in the bracket and see that pendulum hangs vertically.
 - (4) Adjust the eye-piece of the telescope so that the cross wires are clearly visible through it. Focus the telescope on the bar from about one metre distance and see that the point of intersection of the crosswire coincides with the line AB.

(ii) From the line $A'B'C'D'E'$

$$i_1 = \frac{A'C' + C'E'}{2}$$

$$i_2 = \frac{B'C' + C'D'}{2}$$

$$L = i_1 + i_2 = 60$$

$$T = 1.55 \text{ sec}$$

$$g_2 = \frac{4\pi^2 L}{T^2} = 985.9 \text{ cm s}^{-2}$$

$$K_2 = \sqrt{i_1 i_2} = 126.4 \text{ cm}$$

Acceleration due to gravity

$$g = \frac{g_1 + g_2}{2} = \frac{1927.6 \text{ cm s}^{-2}}{2} = 963.8 \text{ cm s}^{-2}$$

Radius of gyration

$$K = \frac{K_1 + K_2}{2} = 121.89 \text{ cm}$$

Also $K = \frac{pQ}{2}$ or the mean length corresponding to the minimum time period.

- (5) Displace the bar slightly and release it. It will begin to vibrate. Note the time for 20 vibrations and also measure the length from the point G upto the bottom of the first hole.
- (6) Suspend the pendulum on the knife edge on B side of the pendulum and take the observation as in step-5 above.
- (7) Now insert the knife edges in holes, 2, 3, ... on either side of the C.G and note the time for 20 vibrations in each case and measure the corresponding distances from G.
- (8) Plot a curve between distance from C.G and the corresponding time period T.
- (9) Draw two horizontal lines ABCDE and A'B'C'D'E and do the calculations.

* Result : (i) The acceleration due to gravity $g = \cancel{9.67} \text{ ms}^{-2}$
Actual value $= 9.8 \text{ ms}^{-2}$
% Error $= \pm 1.73$

* Sources of Error and Precautions :

- (1) The knife edges should be sharp.
- (2) The support should not be jerky.
- (3) Smooth and good sized curve should be drawn.
- (4) Amplitude of vibrations should be small ($< 5^\circ$)